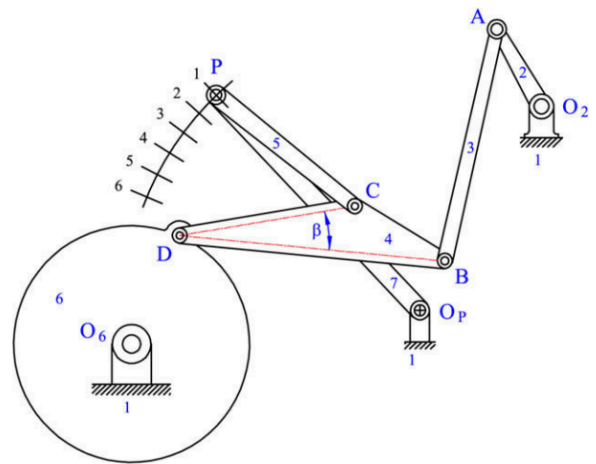


Synthesis of an Adjustable Six-Bar Linkage

This paper investigates the kinematics of an adjustable six-bar linkage where the rotation of the input crank is converted into the oscillation of the output link. This single-degree-of freedom planar linkage is used as a variable-speed transmission mechanism where the input crank rotates at a constant speed and the output link consists of an overrunning clutch mounted on the output shaft. The analysis uses a novel technique in which kinematic coefficients are obtained with respect to an independent variable. Then kinematic inversion is used to express the kinematic coefficients with respect to the input variable of the linkage.

This technique decouples the position equations and provides additional insight into the geometry of the adjustable linkage. The angle through which the output link oscillates, for each revolution of the input crank, can be adjusted by a control arm. This arm allows a fixed pivot to be temporarily released and moved along a circular arc about a permanent ground pivot.



- Pennock, G. R., & Israr, A. (2009). Kinematic analysis and synthesis of an adjustable six-bar linkage. *Mechanism and Machine Theory*, 44(2), 306-323. [[Link](#)]